



Effect of a shield on the hydraulic and thermal performance of a plate-fin heat sink[☆]

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ARTICLE INFO

Available online 13 January 2009

Keywords:

Electronic cooling
Computational fluid dynamics
Heat sink
Shield

ABSTRACT

The hydraulic and thermal performance of a plate-fin heat sink undergoing cross flow forced convection with the introduction of a shield was investigated. With a CFD numerical method, the influence of fin width, fin height, number of fins and Reynolds number were assessed without and with a shield. A shield that tends to decrease the bypass flow effect has a great influence on the variation of the thermal fluid field and the performance of the heat sink. The results of attaching a shield show that more coolant fluid is forced to flow into the fin-to-fin channel to enhance the heat transfer, increasing the pressure drop; this trend is significant at low Reynolds numbers. The decrease of thermal resistance due to the shield diminishes with increasing fin height, but increasing the width of the fins has a more radical effect. For a shield at a particular Reynolds number, the fin geometry should be selected carefully to fit the demands of enhanced effectiveness of heat transfer and decreased power consumption.

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1. Introduction

For the purposes of enhancing cooling performance and avoiding extra power consumption by fans, we investigated the use of a shield to decrease the bypass effect, and the hydraulic and thermal characteristics of a plate-fin heat sink. We adopted a numerical method to assess the distributions of the flow and thermal fields and the performance of a heat sink with a shield, including the effects of fin width, fin height, number of fins and the velocity of approaching flow.

Among published research on heat sinks for improved cooling of electronic components, Yuan [1] used CFD software FLUENT to study the bypass effect of the cross flow in the fin-to-fin passage of a plate-fin heat sink; he varied the space of the two side walls and top wall, and discussed the relations among flow velocity, pressure drop and thermal resistance. The results showed that the air flow enters the fin-to-fin passage and decelerates with distance downstream, thus influencing the effect of heat transfer. El-Sayed et al. [2] varied the fin height, fin width, inter-fin space, number of fins and the distance of fin tips to the shroud to investigate the performance of a plate-fin heat sink. Their conclusions were that the pressure drop increases with increasing Reynolds number and fin height but decreases with increasing inter-fin space and fin width, that increasing the clearance

between the fin tips to the shroud decreases the mean Nusselt number, and that the mean Nusselt number increases with increasing Reynolds number, fin width and inter-fin space but with decreasing fin height. Prstic and Bar-Cohen [3] placed a heat shield in front of a plate-fin heat sink in a transverse duct flow and analyzed the thermal fluid performance both experimentally and numerically; three configurations tested were fully shrouded, partially shrouded and shield conditions. The superior thermal performance achieved in the fully shrouded configuration is accompanied by a penalty of a significant pressure drop. The use of a heat shield can decrease the influence of the bypass effect and induce additional fluid to enter the fin-to-fin channel of a plate-fin heat sink. The thermal performance of this heat shield configuration is near that of a fully shrouded configuration, but its pressure drop is markedly decreased. Elshafei [4] assessed the thermal fluid performance of a plate-fin heat sink under cross flow conditions, both experimentally and theoretically, through parameters for the mainstream velocity, fin density and clearance between tip and shroud. Their effects on the bypass flow, the pressure drop and the thermal performance were discussed. The mean heat transfer coefficient decreased with increasing clearance between tip and shroud, but the effect of this clearance on the mean heat transfer coefficient decreased with increasing Reynolds number. Jonsson and Palm [5] studied experimentally in a wind tunnel the thermal performance of plate-fin heat sinks and strip-fin heat sinks with inline and staggered arrays; they assessed the effects of fin height and bypass conditions on the thermal and hydraulic performance of various heat sinks. The thermal resistance increased with increasing height and width of the wind tunnel duct. In particular, the height of the wind tunnel duct contributed seriously to the effectiveness of heat transfer. By both experiments and one-dimensional modeling,

[☆] Communicated by W.J. Minkowycz.

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Nomenclature

D	Hydraulic diameter, m
E	Total energy per unit mass, J/kg
H	Fin height, m
h	Specific enthalpy, J/kg
k	Thermal conductivity, W/mK
k_{eff}	Effective thermal conductivity, W/mK
n	Number of fins
p	Static pressure, Pa
Q	Overall rate of heat transfer, W
R_{th}	Thermal resistance, °C/W
Re	Reynolds number
S_h	Energy generation rate per unit volume, W/m ³
T	Temperature, K
t	Time, s
u_i	Velocity component, m/s
V	Approaching flow velocity, m/s
W_f	Width of a fin, m
x_i	Coordinates, m

Greek symbols

δ_{ij}	Kronecker delta
ρ	Density, kg/m ³
μ	Dynamic viscosity, Ns/m ²

Dogruoz et al. [6] investigated the hydraulic resistance and thermal performance of in-line square pin-fin heat sinks. The predictions based on the two-branch bypass model agreed satisfactorily with experimental data at large velocities of approach for tall heat sinks, but the discrepancy increased with decreasing approach velocity and height of the heat sink.

Prstic and Bar-Cohen [3] demonstrated the use of a thin sheet-metal heat shield can substantially enhance the thermal performance of a heat sink, becoming comparable to a fully shrouded heat sink but with a pressure drop decreased by about half. As the approach velocities, bypass effects and configuration dimensions of a heat sink system can individually alter the overall thermal performance, we treated the thermal and hydraulic performance with and without the presence of an additional shield by varying approach velocities, fin widths, fin heights and number of fins in a numerical CFD manner.

2. Analysis

The geometry of the theoretical model and boundary conditions are illustrated in Figs. 1 and 2. The dimensions of the theoretical model are 3000 mm×90 mm×200 mm (X×Y×Z). Through a symmetric geometry, a real calculation field of 3000 mm×90 mm×100 mm is used to save computation duration. The thermal and flow fields were calculated numerically with commercial CFD software FLUENT 6.1 [7], according to the following assumptions: the system is in a steady state; the flow is incompressible and turbulent; the fluid and the solid properties are constant, and effects of gravitation and thermal radiation are neglected. The equations governing the fluid-state, with a shield installed on a plate-fin heat sink with cross flow, are the Reynolds-averaged Navier-Stokes equations and the energy equation. The general forms are expressed as

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i} (\rho u_i) = 0 \quad (1)$$

$$\frac{\partial}{\partial t} (\rho u_i) + \frac{\partial}{\partial x_j} (\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \frac{\partial u_l}{\partial x_l} \right) \right] + \frac{\partial}{\partial x_j} (-\rho \overline{u_i' u_j'}) \quad (2)$$

$$\frac{\partial}{\partial t} (\rho E) + \frac{\partial}{\partial x_i} [u_i (\rho E + p)] = \frac{\partial}{\partial x_j} \left[(k_{\text{eff}}) \frac{\partial T}{\partial x_j} + u_i (\tau_{ij})_{\text{eff}} \right] + S_h \quad (3)$$

in which $-\rho \overline{u_i' u_j'}$ is the Reynolds stresses and $(\tau_{ij})_{\text{eff}}$ is the deviatoric stress tensor.

As a result of use of the RNG κ - ϵ turbulent model, the turbulent flow field at lower Reynolds numbers is predictable with increased accuracy and the vortex phenomena with increased precision. We utilized a RNG κ - ϵ turbulent model to solve the complex turbulent thermal flow field and non-equilibrium wall functions to fit the wall boundary conditions [7].

The equation governing the solid-state is the heat conduction equation. Its general form is expressible as

$$\frac{\partial}{\partial t} (\rho h) = \frac{\partial}{\partial x_i} \left(k \frac{\partial T}{\partial x_i} \right) + S_h \quad (4)$$

The overall rate of heat transfer Q of the heat sink is defined as

$$Q = U_h A (T_{\text{ave}} - T_{\infty}) \quad (5)$$

in which U_h denotes the overall heat transfer coefficient, A denotes the total area of heat transfer of the heat sink, T_{ave} denotes the mean temperature of the base of the heat sink, and T_{∞} denotes the temperature of the approaching fluid. The overall heat transfer coefficient is separable into a convection heat transfer coefficient, h_c , and a fin efficiency, η_f , as

$$U_h = h_c \eta_f \quad (6)$$

To compare the effectiveness of heat transfer in the heat sinks, we chose the thermal resistance, R_{th} , as a base parameter for evaluation, defined as

$$R_{\text{th}} = \frac{T_{\text{ave}} - T_{\infty}}{Q} = \frac{1}{U_h A} \quad (7)$$

The Reynolds number is defined as

$$Re = \frac{\rho V D}{\mu} \quad (8)$$

We treated the pressure term with a standard approach, but solved the pressure-velocity coupling with the SIMPLE algorithm. The convergence criterion to terminate the computation was defined as 10^{-7} for the scaled residual of the energy equation or 10^{-5} for the scaled residual of other equations.

3. Results and discussion

The physical model of the plate-fin heat sink is depicted in Fig. 3; its base has length 77 mm, width 70 mm and thickness 8 mm. We investigated the effects of the presence of a shield for varied Reynolds number (Re), fin width (W_f), fin height (H) and number of fins (n), and discuss the dependence of the thermal performance of the heat sink on those parameters. We tested four Reynolds numbers – Re=10,000, 20,000, 30,000 and 40,000, five fin widths – W_f =0.5, 1.0, 1.5, 2.0 and 2.5 mm, five fin heights – H=15, 30, 45, 60 and 75 mm, and four numbers of fins – n=10, 20, 30 and 40. The power of the heater at the bottom of the heat sink was maintained at 20 W.

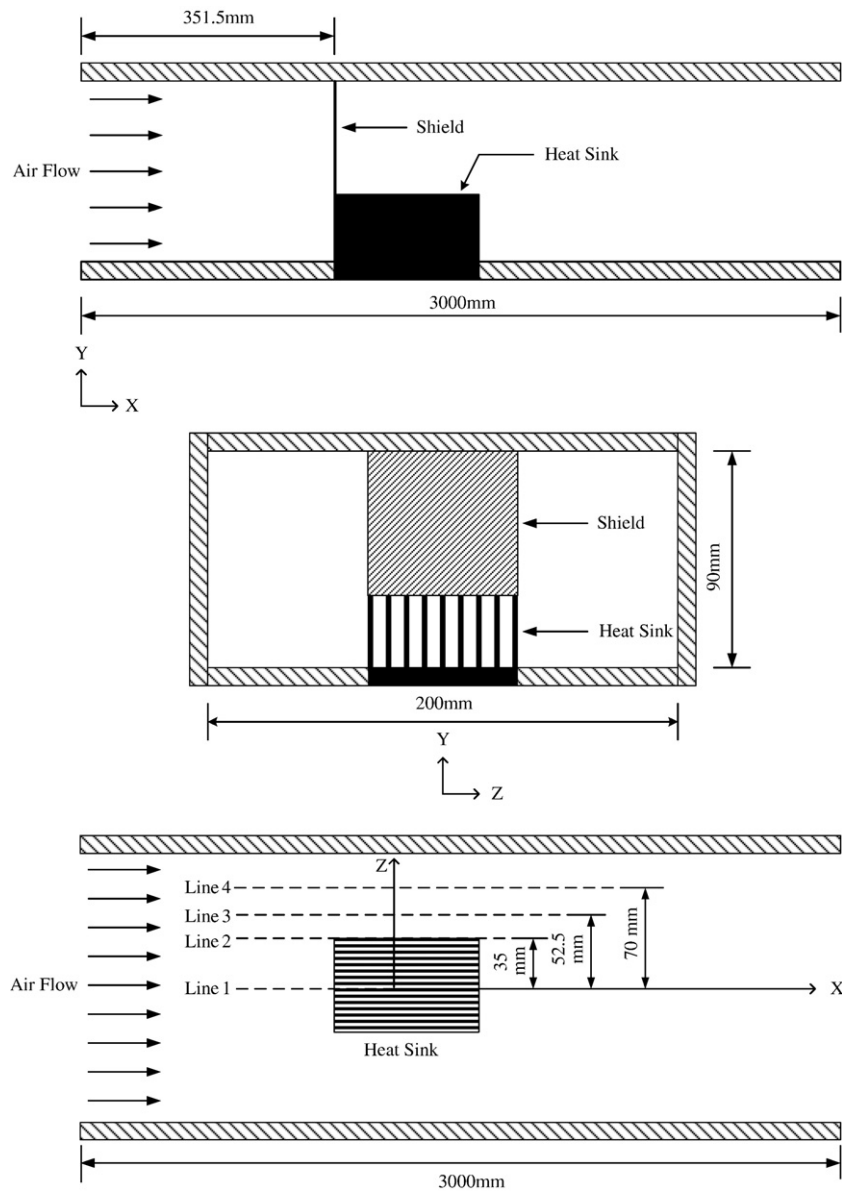


Fig. 1. Illustration of the attachment of a shield to a plate-fin heat sink.

The applied grid system was certified under conditions of a heat sink with $W_f = 1.5$ mm and $H = 45$ mm, with a shield and at $Re = 10,000$. Three densities of grids used were 140,434, 248,600 and 347,496. The computational result of the mean surface temperature of the heat sink varied within 0.14%, with 4.2% deviation deduced for thermal resistance, for a number of grids increased from 140,434 to 347,496. The computational result of the mean surface temperature of the heat sink varied within 0.003%, with a deduced 0.1% deviation in thermal resistance, for the number of grids increased from 248,600 to 347,496. To save computing time without a penalty of poor accuracy, we applied a system of grids numbering 248,600 for this work.

3.1. Comparison of numerical simulation and experiment

The confidence of numerical simulation was verified with experiments [8], from which three heat sinks with $W_f = 1.5$ mm and $H = 30, 45$ and 60 mm, without a shield, at various Reynolds numbers were chosen for comparison. Fig. 4 shows that the average discrepancy of thermal resistance between numerical simulation and experimental data is 17.5%. Because this agreement is satisfactory, we investigated

the behavior of hydraulic and thermal characteristics by numerical simulation.

3.2. Pressure distribution

To understand the influence on hydraulic performance of the shield and various parameters, we selected pressure distributions of four lines (Line 1 ~ Line 4) on the bottom plate along the flow stream, as depicted in Fig. 1, for further examination. The pressure distributions for a heat sink of dimensions $W_f = 1.5$ mm and $H = 45$ mm, without and with a shield, at $Re = 10,000$ and 30,000, are depicted in Figs. 5 and 6, respectively. As a result of the blockage by the heat sink, the pressures evidently increase at a central line in front of the heat sink (Line 1) as the flow approaches it. Part of the fluid enters the fin-to-fin channels to perform heat transfer, and the other part accelerates from the sides of the heat sink and causes a pressure decrease, which is most evident at the side of the heat sink (Line 2). As the fluid enters the heat sink, the pressure at the centerline of the heat sink (Line 1) decreases significantly, but rises abruptly at the edge of the heat sink (Line 2). After the flows pass the heat sink, the pressures attain a

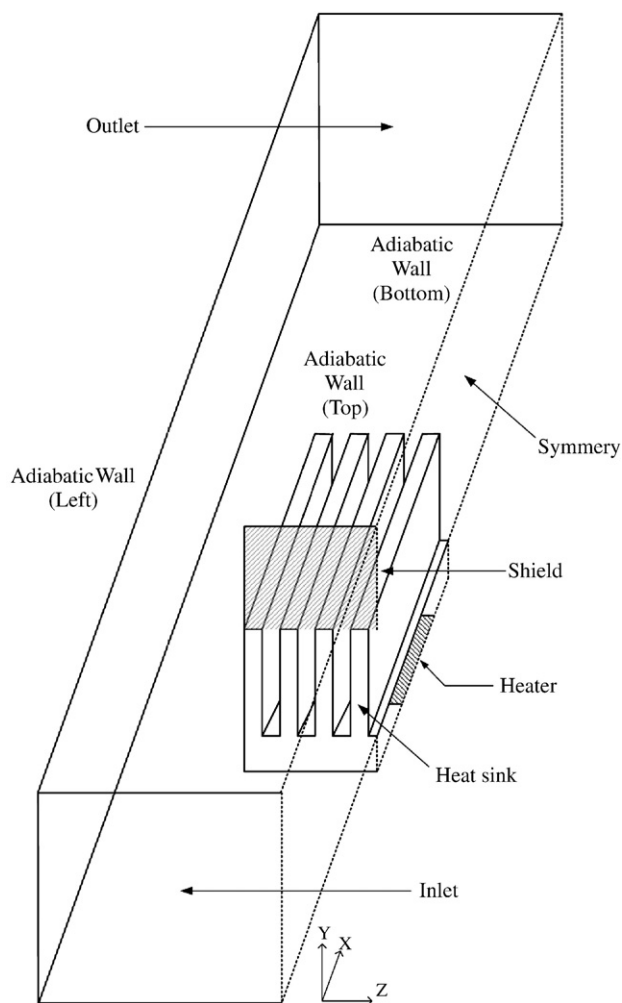


Fig. 2. Settings of boundary conditions.

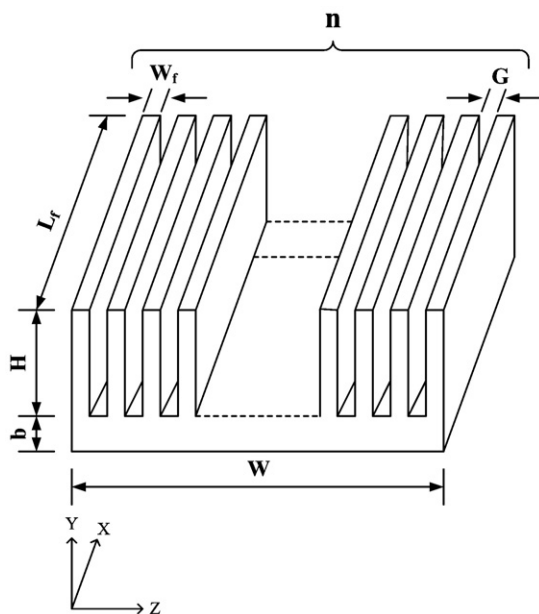


Fig. 3. Geometry of a plate-fin heat sink.

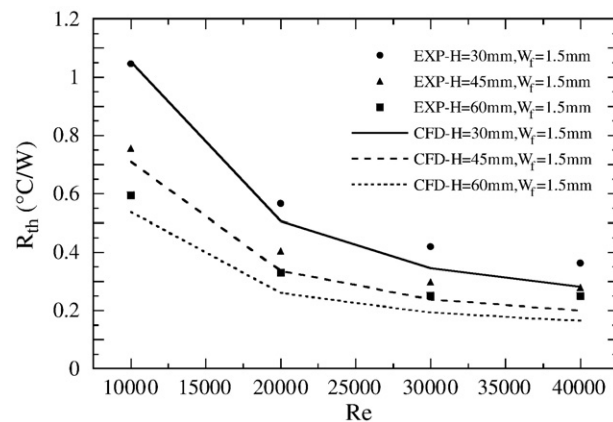
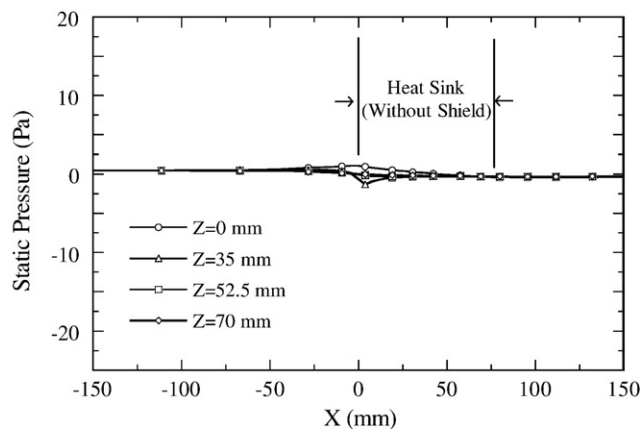


Fig. 4. Comparison of simulation and experimental data with no shield attached, $W_f = 1.5$ mm, and $H = 30, 45$ and 60 mm.

steady state. Both an increased Reynolds number and the existence of a shield greatly augment the change of pressure distribution. The presence of a shield causes a greatly increased pressure upstream of the heat sink and increases the pressure drop across the heat sink for Line 1.

(a) $Re = 10000$



(b) $Re = 30000$

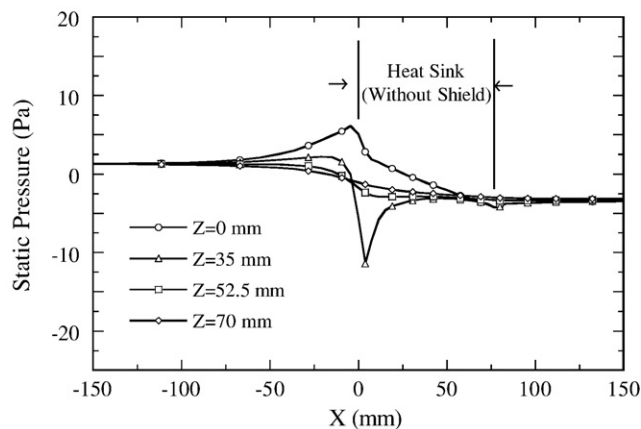


Fig. 5. Pressure distributions for a heat sink ($W_f = 1.5$ mm and $H = 45$ mm) without a shield.

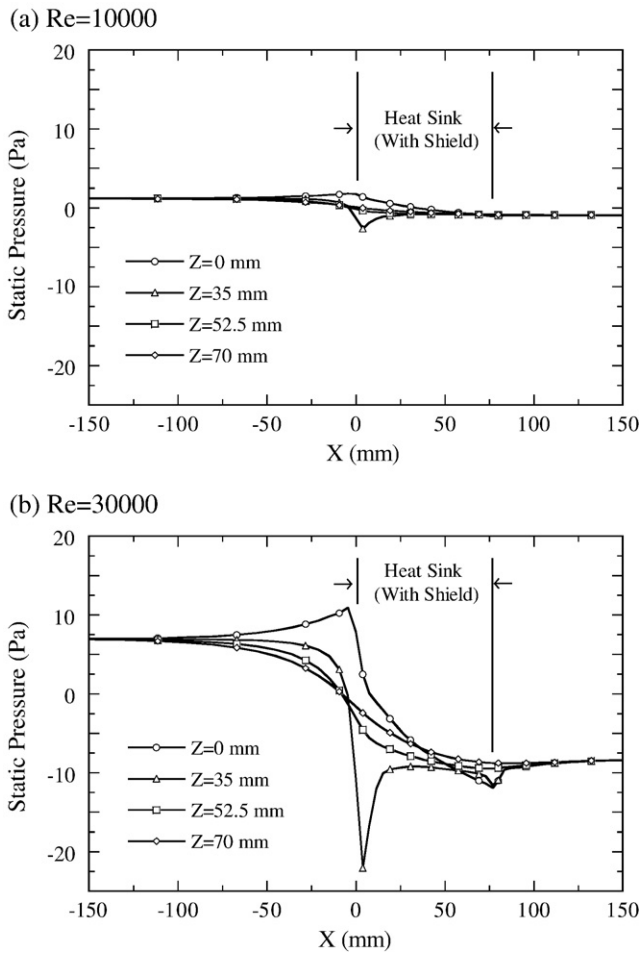


Fig. 6. Pressure distributions for a heat sink ($W_f=1.5$ mm and $H=45$ mm) with a shield.

3.3. Velocity distribution

The velocity distributions on the symmetric plane of the heat sink, which has $W_f=1.5$ mm and $H=45$ mm, without and with a shield, at $Re=10,000$ and $30,000$, are depicted in Figs. 7 and 8, respectively. Increasing the Reynolds number increases the maximum velocity. In the case of no shield, the fluid decelerates as the flow enters the fin-to-fin channel of the heat sink; maximum velocities occur at the upper parts of the heat sink. Only part of the fluid hence enters the fin-to-fin channel to perform the force convection; most coolant fluid bypasses through the top side of the heat sink. Flow recirculation occurs downstream of the heat sink, but becomes ambiguous with increasing Reynolds number. When a shield is present, the flow entering the fin-to-fin channel accelerates; maximum velocities occur in the fin-to-fin channel, resulting from the blockage effect of a shield, which forces more coolant fluid to enter the fin-to-fin channel and increases the velocity in this space. The presence of a shield thus bestows an advantage of enhancing the heat transfer from the heat sink. Flow recirculation occurs in the flow field behind the shield, and moves downstream along the duct direction as the Reynolds number increases.

3.4. The influence of Reynolds number and a shield on thermal resistance

Fig. 9 illustrates the thermal resistances of the heat sinks, with fin width $W_f=1.5$ mm and height $H=15, 30$ and 60 mm, with and without a shield, at various Reynolds numbers. The thermal resistance

decreases substantially, and diminishes smoothly with increasing Reynolds number whether a shield was installed or not. For instance, the thermal resistance of the heat sink of fin height $H=30$ mm, without shield, at Reynolds number $Re=10,000$, is 1.05 °C/W, decreasing to 0.51 °C/W with Reynolds number increased to $20,000$, a decrease by 51.4% . On further increase of Reynolds number from $20,000$ to $30,000$, the decrease is 31.3% , and a further decrease by 20% as the Reynolds number increases from $30,000$ to $40,000$. The thermal resistance thus decreases but this trend slows with increasing Reynolds number. The thermal resistance decreases also as the fin height increases, with or without a shield.

3.5. The influence of fin height and a shield on thermal resistance

Fig. 10 depicts the effect of fin heights ($H=15, 30, 45, 60$ and 75 mm) and a shield on the thermal resistances of heat sinks, with constant fin width $W_f=1.5$ mm, at varied Reynolds numbers. The deviation of thermal resistances for fin heights $H=15$ and 75 mm, both without a shield, is 1.52 °C/W at $Re=10,000$ and only 0.41 °C/W at $Re=40,000$. An increased fin height hence increases the area of heat transfer and thus enhances the thermal effectiveness. The decline of thermal resistance due to increased fin height decreases with increasing Reynolds number. The thermal resistances at fin heights 60 and 75 mm are almost the same for Reynolds number $40,000$. The variation of fin height at low Reynolds numbers has a greater impact on the thermal performance than at high Reynolds numbers.

With a shield, an increased fin height decreases the height of the shield because the duct height is constant, and this decreased area of the shield causes decreased blockage and has less impact on the thermal resistance. The effect of fin height with a shield is evident at low Reynolds number but almost vanishes for a fin height greater than 60 mm at Reynolds number $40,000$. Practical fin heights with shields at various Reynolds numbers should be thus chosen to fulfill the requirements of satisfactory thermal performance and modest cost.

3.6. The influence of fin width and a shield on thermal resistance

Fig. 11 depicts the effect of fin widths ($W_f=0.5, 1.0, 1.5, 2.0$ and 2.5 mm) and a shield on the thermal resistances of heat sinks, of constant fin height $H=45$ mm, at varied Reynolds numbers. Heat sinks without a shield show greater thermal resistance than with a shield at Reynolds number $10,000$, and the thermal resistance increases with increasing fin width. The latter condition might result from the increasing fin width decreasing the width of the fin-to-fin channel, which, in turn, causes less coolant fluid to enter the channel and diminish the thermal performance. The total area of heat transfer also increases as the fin width increases. The combination of these effects causes a balanced status, and a fin width optimized for least thermal resistance exists. The optimal fin width increases with increasing Reynolds number because the coolant fluid in the fin-to-fin channel accelerates and enhances heat transfer. The optimal fin widths are $0.5, 1.0, 1.5$ and 1.5 mm at $Re=10,000, 20,000, 30,000$ and $40,000$, respectively, whether a shield is installed or not. A suitable fin width can thus be selected to achieve optimal thermal performance at a particular velocity of coolant flow.

When a shield is installed, the blockage effect forces more fluid to enter the fin-to-fin channel to perform heat transfer and thus to decrease the thermal resistance. The effect of a shield is more significant with increasing fin width of the heat sink. For $Re=20,000$ and $W_f=0.5, 1.5$ and 2.5 mm, the use of a shield causes $9.4\%, 26.5\%$ and 30.2% decreased thermal resistance, respectively. The effect of a shield on thermal performance of a heat sink is hence more significant as the fin width increases. Both increasing Reynolds number and adding a shield can enhance the thermal performance, but this influence is insignificant for a small fin width. This trend might result because the heat sink at a small fin width has a wide fin-to-fin space and a small

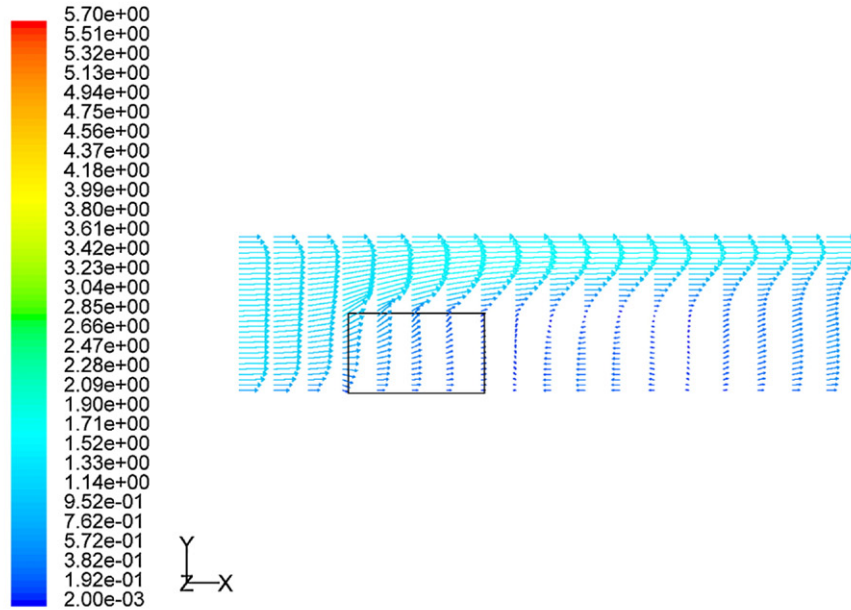
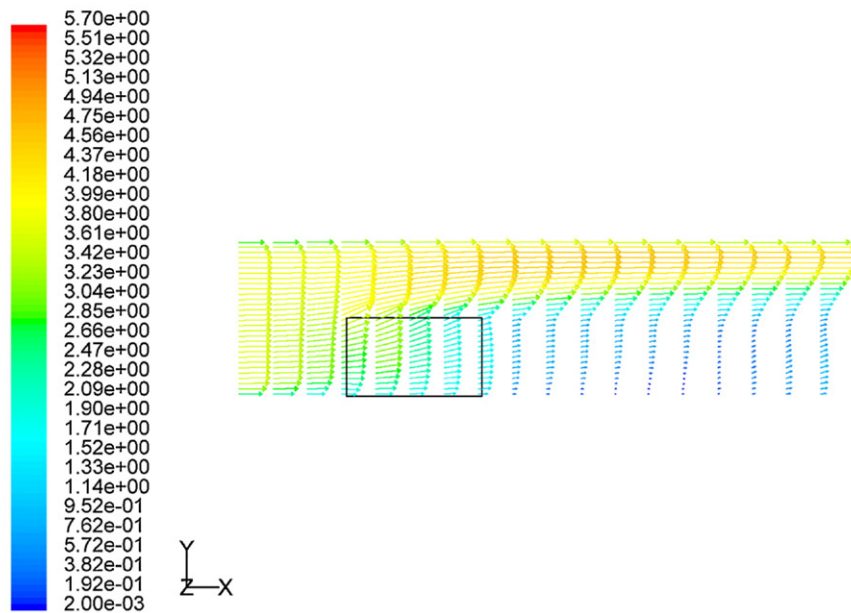
(a) $Re=10000$ (b) $Re=30000$ 

Fig. 7. Velocity distributions on the symmetric plane of a heat sink ($W_f=1.5$ mm and $H=45$ mm) without a shield.

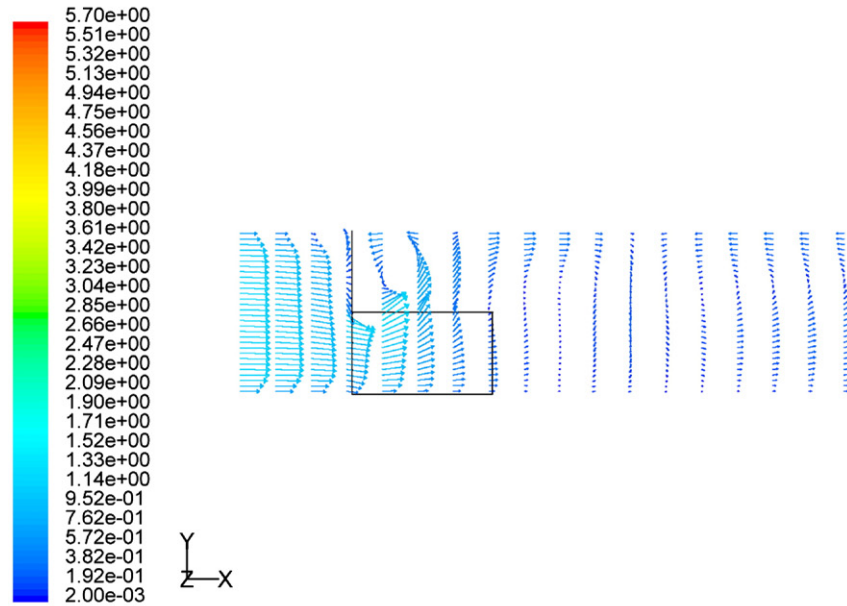
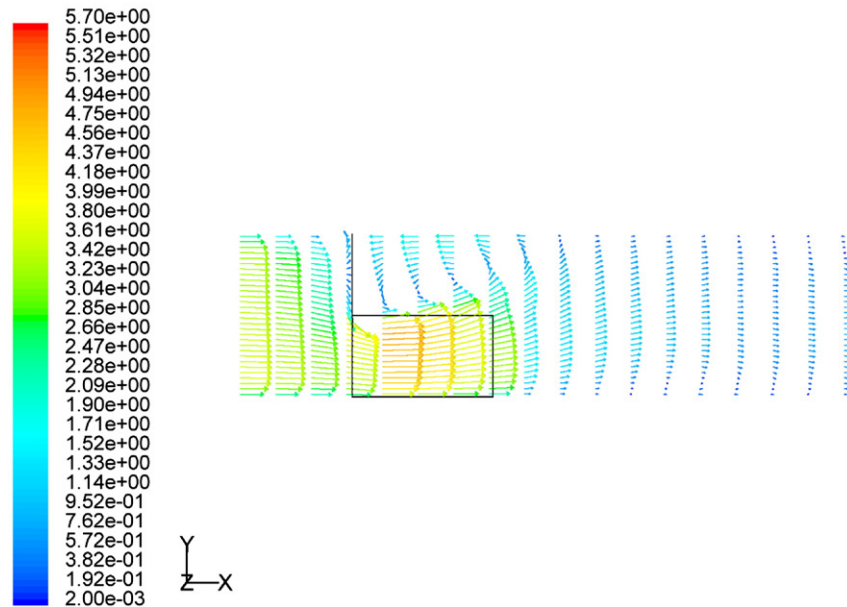
flow resistance, so that the flow can enter the fine-to-fin channel without much restraint, and the fluid squeezed by the blockage of a shield into the fin-to-fin channel is not greatly increased.

3.7. The influence of number of fins and a shield on thermal resistance

To enable a comparison of the effect of the number of fins on thermal performance of a heat sink, the total volume of all fins was maintained constant. The widths of the fins hence decrease with increasing number of fins. Fig. 12 depicts the dependence of thermal resistance on number of fins ($n=10, 20, 30$ and 40) of the plate-fin heat sinks, with or without a shield, and with fin height constant at $H=45$ mm. The thermal resistance increases with increasing number of fins at $Re=10,000$ and without a shield, perhaps because the flow resistance is least at number of fins

$n=10$. The coolant flow most directly enters the fin-to-fin channel to carry heat away, and the flow resistance dominates the thermal performance. The least thermal resistance shifts toward number of fins $n=20$ as the Reynolds number increases to 20,000. Further increasing the number of fins increases the thermal resistance. The increasing number of fins increases the total area of heat transfer that is beneficial for decreasing the thermal resistance, but concurrently increases the flow resistance so as to restrict the coolant flow entering the fin-to-fin channel to carry heat away. These two opposing effects combat and compensate; as a result, an optimal number of fins exists. At Reynolds number 40,000, the influence of the number of fins on thermal resistance becomes ambiguous for fins numbering over 20.

From a comparison of the results with a shield at $Re=10,000$, the least thermal resistance appears for $n=20$. The optimal choice of

(a) $Re=10000$ (b) $Re=30000$ Fig. 8. Velocity distributions on the symmetric plane of a heat sink ($W_f=1.5$ mm and $H=45$ mm) with a shield.

number of fins increases when the shield is present, because more coolant fluid is squeezed into the fin-to-fin channel through the blockage effect of the shield. At $Re=40,000$, the optimal number of fins becomes $n=30$, but the difference of thermal resistance with $n=20$ is small. On grounds of difficulty and time consumption of machining and of cost, the choice $n=20$ is preferable. Overall, the presence of a shield decreases the thermal resistance at various Reynolds numbers, but the effect diminishes with increasing Reynolds number.

4. Conclusions

According to our numerical investigation of the effect of a shield on the thermal performance of a plate-fin heat sink undergoing cross flow, we proffer the following conclusions.

1. The pressure drop caused by the presence of a heat sink increases with increasing Reynolds number and the attachment of a shield reinforces this tendency.
2. The attachment of a shield to a heat sink squeezes more coolant fluid into the fin-to-fin channel and enhances the thermal performance.
3. Thermal resistance decreases with increasing Reynolds number, but the effect diminishes at high Reynolds numbers. The enhancement of thermal performance by a shield is much greater at low Reynolds numbers.
4. Increasing the fin height enhances the performance of the heat sink, but the effectiveness decreases with increasing Reynolds number. Increasing the fin height also decreases the height of a shield, in turn decreasing the impact of blockage on thermal resistance.

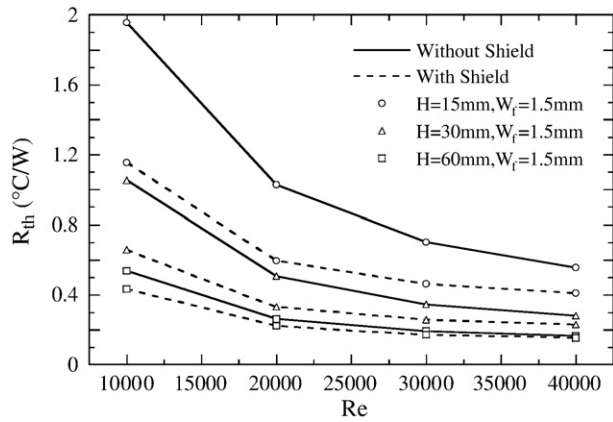


Fig. 9. Influence of Reynolds number and a shield on thermal resistance.

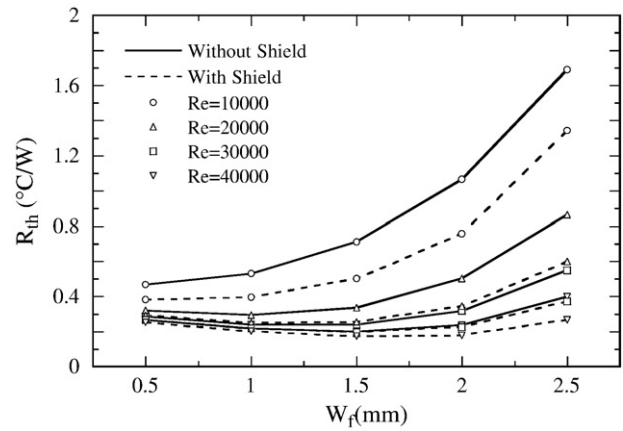


Fig. 11. Influence of fin width and a shield on thermal resistance, $H=45$ mm.

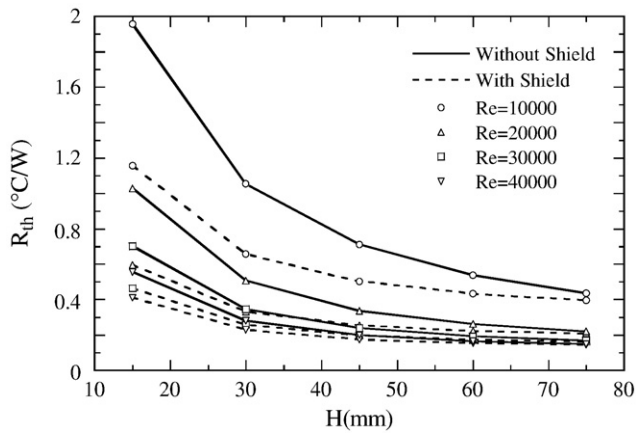


Fig. 10. Influence of fin height and a shield on thermal resistance, $W_f=1.5$ mm.

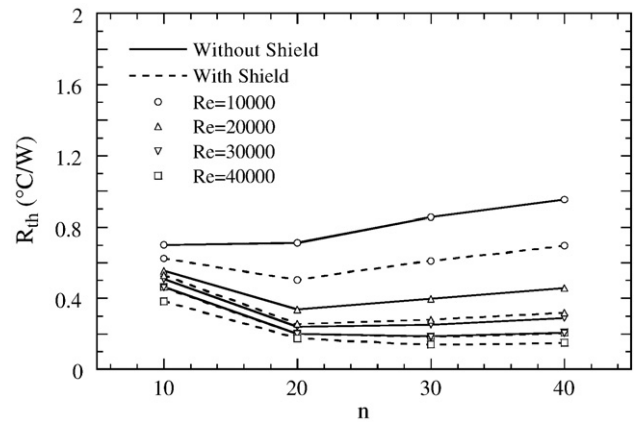


Fig. 12. Influence of number of fins and a shield on thermal resistance, $H=45$ mm.

5. The optimal fin width, i.e., that of least thermal resistance, increases with increasing Reynolds number. The influence of a shield is much greater for a large fin width.
6. The optimal number of fins increases with increasing Reynolds number, and is greater with a shield at low Reynolds numbers.

Acknowledgement

The authors would like to thank the National Science Council of the Republic of China for financially supporting this research under Contract No. NSC 96-2221-E-211-017.

References

- [1] T.D. Yuan, Computational modeling of flow bypass effects on straight fin heat sink in a rectangular duct, Twelfth IEEE Semi-therm Symposium (1996) 164–168.

- [2] S.A. El-Sayed, S.M. Mohamed, A.M. Abdel-latif, A.E. Abouda, Investigation of turbulent heat transfer and fluid flow in longitudinal rectangular-fin arrays of different geometries and shrouded fin array, *Exp. Therm. Fluid Sci.* 26 (8) (2002) 879–900.
- [3] S. Prstic, A. Bar-Cohen, Heat shield—an enhancement device for an unshrouded, forced convection heat sink, *J. Electron. Packag.* 128 (2) (2006) 172–176.
- [4] E.A.M. Elshafei, Effect of flow bypass on the performance of a shrouded longitudinal fin array, *Appl. Therm. Eng.* 27 (13) (2007) 2233–2242.
- [5] H. Jonsson, B. Palm, Thermal and hydraulic behavior of fin and strip fin heat sinks under varying bypass conditions, *IEEE Trans. Compon. Packag. Technol.* 23 (1) (2000) 47–54.
- [6] M.B. Dogruoz, M. Urdaneta, A. Ortega, Experiments and modeling of the hydraulic resistance and heat transfer of in-line square pin fin heat sinks with top by-pass flow, *Int. J. Heat Mass Transfer* 48 (23–24) (2005) 5058–5071.
- [7] Fluent 6.1 user's guide, Fluent Inc., New Hampshire, USA, 2003.
- [8] H. Y. Li, S. M. Chao, Performance measurement of plate-fin heat sinks with cross flow cooling, *Int. J. Heat Mass Transfer*, submitted for publication.